

SQL Technical Submission

Task 2: Chinook Database (Music Store Data Extraction & Analytics)

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Role:	Data Analyst Intern
Assignment:	Week 3 - Relational Schema Analytics (Chinook Environment)
Environment:	MySQL / PostgreSQL (Schema: `chinook`)

1. Executive Summary & Schema Relationships

This report outlines the data analysis performed on the **Chinook Relational Database** representing a digital media store. Unlike flat data structures, the Chinook schema features robust entity-relationship networks, utilizing Primary Keys (PK) and Foreign Keys (FK) across standard corporate entities, including customers, employees, invoices, tracks, and genre catalogs.

2. Core SQL Queries & Business Implementations

Task 2.1: Basic Filtering & Revenue Sorting

Isolates invoices originating from a target market (e.g., 'USA') and sorts the transactions to find high-value invoice items.

```
SELECT InvoiceId, CustomerId, InvoiceDate, BillingCity, Total FROM Invoice WHERE BillingCountry = 'USA' ORDER BY Total DESC;
```

Strategic Business Insight: Direct geographic constraints enable marketing managers to identify top purchasing locations within major regions to launch seasonal regional campaigns.

Task 2.2: Aggregations & Grouped Sales Distributions

Computes total revenue sums, average billing counts, and active order distributions grouped by specific operational territories.

```
SELECT BillingCountry, SUM(Total) AS Total_Revenue, AVG(Total) AS Average_Invoice_Value, COUNT(InvoiceId) AS Total_Invoices FROM Invoice GROUP BY BillingCountry ORDER BY Total_Revenue DESC;
```

Strategic Business Insight: Grouped revenue distributions give executive teams absolute clarity into market size. This helps determine which international regions drive company profitability.

Task 2.3: Targeted Structural Group Screening using HAVING

Filters out historical customers whose absolute collective purchasing value passes beyond a significant threshold benchmark (e.g., \$40 total spent).

```
SELECT CustomerId, SUM(Total) AS Cumulative_Spent FROM Invoice GROUP BY CustomerId HAVING Cumulative_Spent > 40 ORDER BY Cumulative_Spent DESC;
```

Strategic Business Insight: Applying HAVING conditions helps identify VIP customer groups. This allows the business to offer personalized loyalty rewards or early access to new music releases.

3. Advanced Relational Databases & Multi-Table Joins

Task 3.1: Multi-Table Relational Join Implementation

Connects transaction rows across three tables (Invoice, InvoiceLine, and Track) to match invoice lines with explicit audio track identifiers.

```
SELECT i.InvoiceId, c.FirstName, c.LastName, t.Name AS TrackName, il.UnitPrice, il.Quantity FROM Invoice i INNER JOIN Customer c ON i.CustomerId = c.CustomerId INNER JOIN InvoiceLine il ON i.InvoiceId = il.InvoiceId INNER JOIN Track t ON il.TrackId = t.TrackId LIMIT 10;
```

Strategic Business Insight: Joining customer records with exact track names provides clear visibility into specific digital content demand. It shows exactly which musical tracks drive standard purchase baskets.

Task 3.2: Subquery Anomaly Identifiers

Extracts individual billing items holding values larger than the general global average billing price calculated across all time.

```
SELECT InvoiceId, CustomerId, Total FROM Invoice WHERE Total > (SELECT AVG(Total) FROM Invoice) ORDER BY Total DESC;
```

Strategic Business Insight: Automatically filters out routine transactions, flagging larger order metrics for auditing, financial forecasting, and account expansion tracking.

Task 3.3: Chronological Segment Distribution (Window Functions)

Deploys ROW_NUMBER() and RANK() metrics over localized tracking partitions (e.g., Billing Country) to sequence track sizes smoothly.

```
SELECT BillingCountry, InvoiceId, Total, ROW_NUMBER() OVER (PARTITION BY BillingCountry ORDER BY Total DESC) AS Row_Num, RANK() OVER (PARTITION BY BillingCountry ORDER BY Total DESC) AS Revenue_Rank FROM Invoice;
```

Strategic Business Insight: Analytical partitioning helps managers rank performance locally within individual countries. This keeps all raw invoice line details clear and uncompressed for complete auditing transparency.

4. Target Optimizations & Clean SQL Architecture

Indexes were deployed across heavily queried field parameters (like `CustomerId` inside target search tables) to speed up performance and prevent engine execution bottlenecks.

```
CREATE INDEX idx_chinook_cust ON Invoice(CustomerId);
```

Strategic Business Insight: Indexing heavily queried foreign key relationships prevents slow full-table scans. This keeps reporting tools fast and responsive as data volume grows.